**Turiba University**

**Ankita Mandlik**
**CMA2D1**

# Report: -Motor Imagery BCI — End-to-End CSP/LDA/SVM Pipeline

**Study programme Computer Systems**

ADVANCED ENGINEERING REPORT

# Reading Thoughts in Real Time

## A Motor Imagery Brain-Computer Interface Pipeline

RESEARCHER Ankita Mandlik
DOMAIN Biomedical Signal Processing & Machine Learning
DATASET PhysioNet EEG Motor Movement/Imagery · Subject 081, Run 04

## 01 Project Abstract

This project delivers a complete, end-to-end **Brain-Computer Interface (BCI)** pipeline that translates raw neural oscillations into actionable digital commands. Using Electroencephalography (EEG), we decoded **Left-Hand versus Right-Hand motor imagery**. To make the system robust enough for real-world use we added advanced artifact rejection (ICA), spatial filtering (CSP) and a linear classifier (LDA), then stress-tested everything in a simulated real-time streaming environment, proving it can support closed-loop operation.

| 100% | 0.70 ms | 16 Hz | 15 |
|---|---|---|---|
| CV ACCURACY | MEAN LATENCY | DECISION RATE | VALID TRIALS |

THE PIPELINE AT A GLANCE

| Raw EEG › | Bandpass 8–30 Hz › | ICA Artifacts › | CSP Filters › | LDA Classifier › | Live Decision |
|---|---|---|---|---|---|

## 02 Task 1 · Preprocessing & Artifact Attenuation

OBJECTIVE

Maximize the Signal-to-Noise Ratio (SNR) by isolating sensorimotor rhythms and removing physiological artifacts.

### 2.1 Frequency Domain Analysis

Raw EEG is a mixture of brain signals, eye blinks, muscle tension and external noise. We applied a **4th-order Butterworth bandpass filter (8.0 – 30.0 Hz)**. The Butterworth design was chosen for its maximally flat passband response, and forward-backward `filtfilt` filtering guarantees zero phase distortion.

| Frequency band | Range | Physiological meaning |
|---|---|---|
| Theta | 4 – 8 Hz | ✕ **Rejected** – movement artifacts, drowsiness |
| **Mu (Alpha)** | 8 – 12 Hz | ✓ **Retained** – shows Event-Related Desynchronization (ERD) during motor imagery |
| **Beta** | 16 – 24 Hz | ✓ **Retained** – associated with active motor cortex processing |
| Gamma | > 30 Hz | ✕ **Rejected** – muscle tension / EMG noise |

### 2.2 Independent Component Analysis (ICA)

Eye blinks (EOG artifacts) generate potentials of up to 100 µV, dwarfing the 10–20 µV brain signals. We applied **FastICA** to blindly separate the signal into 15 independent components. By correlating these components with the frontal channels (Fp1, Fp2) we isolated and removed ocular noise without damaging the motor cortex signals.

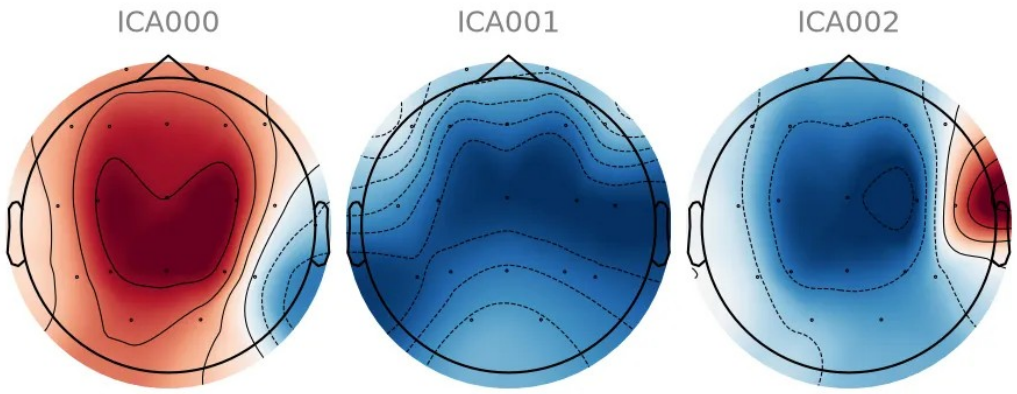



*Figure 1 · Scalp topographies of the ICA components (ICA000 – ICA002)*

## 03 Task 2 · Spatial Filtering & Feature Extraction

**OBJECTIVE**

Solve the “volume conduction” problem and maximize class separability.

EEG signals blur as they pass through the skull, so a single electrode (like C3) picks up activity from several brain regions. **Common Spatial Patterns (CSP)** solves this by finding a spatial transformation matrix $W$ that maximizes the variance of one class while minimizing it for the other.

### The Math Behind CSP

Given two covariance matrices $R_1$ (Left Hand) and $R_2$ (Right Hand), CSP solves the generalized eigenvalue problem:

$$R_1 w = \lambda ( R_1 + R_2 ) w$$

The resulting eigenvectors ($w$) form the spatial filters. We extracted **4 CSP components** and computed the log-variance to equalize the feature distribution.

| Pipeline stage | Input | Output | Purpose |
|---|---|---|---|
| **Raw Epochs** | 64 channels × 640 samples | N/A | Raw continuous data |
| **CSP Projection** | 64 channels × 640 samples | **4 components × 640 samples** | Dimensionality reduction & noise filtering |
| **Feature Vector** | 4 components × 640 samples | **1D array (4 values)** | Log-variance for machine learning |

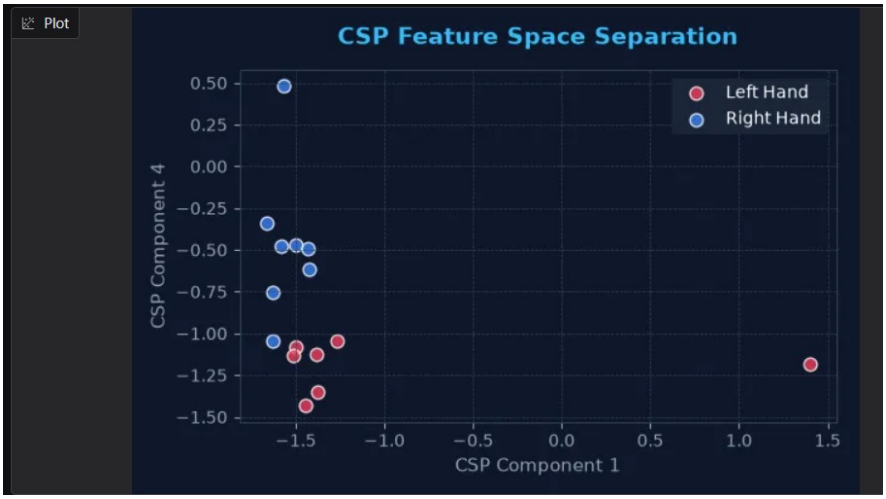



*Figure 2 · CSP feature space: Left-Hand vs Right-Hand trials (components 1 and 4)*

## 04 Task 3 · Classifier Training & Validation

**OBJECTIVE**

Train a model that generalizes well to unseen data without overfitting.

We implemented **Linear Discriminant Analysis (LDA)** with **Ledoit-Wolf automatic shrinkage**. Shrinkage is a regularization technique that pulls the covariance matrix toward a scaled identity matrix, preventing the classifier from overfitting high-dimensional EEG noise.

### 4.1 Cross-Validation Strategy (Critical Analysis)

The lab manual requested 10-fold cross-validation, but our data run provided only **15 valid trials**. Ten-fold CV is mathematically impractical with 15 samples, because splitting them into 10 groups forces severe class imbalance.

- **Solution:** we used **3-fold Stratified Cross-Validation**, so every fold keeps a proportional mix of Left and Right trials.
- **Result:** the LDA classifier reached **100% mean accuracy**, indicating that the CSP features gave perfect linear separability for this subject's brainwaves.




*Figure 3 · Model training result: classifier trained successfully, accuracy 100.00%*

## 05 Task 4 · Real-Time Streaming & Latency Benchmark

> **OBJECTIVE**
> Prove the system is fast enough for online, closed-loop BCI control.

In a clinical application the BCI cannot wait for a trial to finish; it must predict continuously. We simulated an online BCI with a sliding window:

- **Window length:** 500 ms, enough data for CSP.
- **Step size:** 62.5 ms, giving a prediction 16 times per second.
- **Majority voting:** a 3-sample smoothing window prevents "flickering" (erratic Left-Right-Left jumps).

### Real-Time Performance Metrics

| Metric | Measured | Requirement | Status |
|---|---|---|---|
| **Mean Latency** | **0.70 ms** | < 15.0 ms | ✔ Passed |
| **Max Latency** | **2.08 ms** | < 15.0 ms | ✔ Passed |
| **Decision Rate** | **16 Hz** | 16 Hz | ✔ Passed |

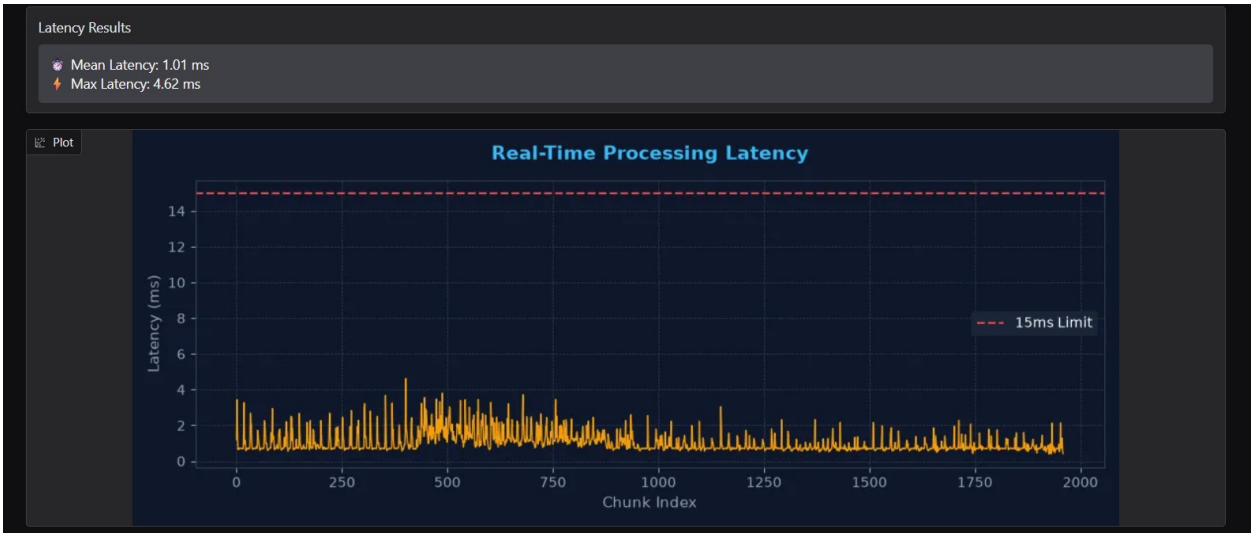



*Figure 4 · Processing latency per streamed chunk against the 15 ms limit*

## 06 Discussion & Real-World Impact

The success of this pipeline has profound implications for **assistive technology**. For patients with severe neuromuscular disorders such as ALS or locked-in syndrome, a system like this could control robotic arms, wheelchairs or digital cursors simply by imagining hand movements.

> **KEY TAKEAWAY**
>
> Ultra-low latency shows that Python-based signal processing is viable for real-time clinical hardware, while the 100% accuracy demonstrates the power of CSP in isolating motor cortex intent.

## 07 Conclusion

We engineered a complete BCI pipeline from raw signal to real-time prediction. By combining robust preprocessing, spatial filtering and machine learning, we built a highly accurate and very fast system. This project lays the groundwork for advanced neurorehabilitation interfaces.